

RAHUL J HIRUR

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PROFESSIONAL SUMMARY

Computational Engineer specializing in Applied AI and Foundation Models, with proven expertise in integrating LLMs and machine learning pipelines into scalable industrial systems, looking for roles in AI and deep learning.

EXPERIENCE

AI Engineer - Intelligent Automation

Qentelli

Mar 2026 – Present

Hyderabad, India

- Engineered a hybrid deduplication pipeline, integrating deep semantic embeddings with lexical similarity metrics to identify and isolate redundant test scenarios.
- Developed LLM-based refinement service to orchestrate automated text processing to standardize and refine test cases into consistent and compliant structures.
- Deployed an end-to-end FastAPI application on Azure, achieving a 30% reduction in total testcase volume and optimizing the testing life cycle.

Master Thesis - AI based High-Precision 3D Reconstruction and Perception

FAPS – Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik

Feb 2025 – Dec 2025

Erlangen, Germany

- Developed a full end-to-end 3D reconstruction Generative (GenAI) pipeline, encompassing image acquisition, processing, and final 3D construction for real-time robotic perception.
- Benchmarked and implemented two distinct depth methodologies: a Stereo Foundation Model (e.g., NVIDIA FoundationStereo) and a Monocular Foundation Model (e.g., DepthAnything) for disparity and depth calculation.
- Implemented Phase Shifting Profilometry based Prompt Depth Anything model for accurate depth calculation.
- Applied advanced computer vision techniques for Depth Segmentation and geometric processing to ensure the generation of accurate, high-density 3D point clouds.

Werkstudent – Data Engineering and Analytics

Robert Bosch GmbH - Vehicle Motion

Apr 2023 – Nov 2025

Abstatt, Germany

- Developed metrics and algorithms to evaluate Electronic Stability Program (ESP) simulation results.
- Implemented a variational autoencoder (VAE) for anomaly detection in simulated data, utilizing data profiling to handle bias.
- Built and deployed a Plausibility Check dashboard using Streamlit to assess ESP test bench results, utilizing Docker and Kubernetes for scalable container orchestration.
- Developed a Python-based Project Assessment tool to evaluate ESP project feasibility based on fatigue and environment conditions; managed deployment via Rancher.
- Migrated the ESP SPoC Tool from MATLAB to Python, streamlining the codebase.

System Interface Engineer

Robert Bosch - BGSW

Jan 2020 - Sep 2022

Bengaluru, India

- Analyzed and validated performance and fatigue loads for Brake Modulation Systems (ESP and IPB).
- Developed MATLAB ETL pipelines to validate ESP simulation data, reducing manual analysis time by 95%.
- Developed a MobileNet-based CNN pipeline with 96% accuracy to analyze 3D CAD drawings and automatically detect and document mechanical design changes.
- Developed and maintained MATLAB-based visualization tool with time-series statistical baselines to monitor valve pressure modulations and detect mechanical failures.
- Engineered Jenkins CI/CD pipelines to automate the ESP simulation workflow, increasing testing efficiency.

SKILLS

Programming Languages	Python, Matlab, ROS, SQL, C
Deep Learning	PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, Scikit-Learn, OpenCV, RAG
Web	Streamlit, Javascript, Dash, Plotly, Power BI
Cloud	Azure, Git, Jenkins, Kubernetes, Docker
Others	Siemens NX, NXOpen, FastAPI, HiTL, Digital Signal Processing, Statistics

EDUCATION

Master of Science (M. Sc.) in Computational Engineering	Oct 2022 – Mar 2026
Friedrich Alexander Universität Erlangen-Nürnberg	<i>Erlangen, Germany</i>
<i>Relevant Coursework:</i> Deep Learning, Computer Vision(CV), Pattern Recognition(NLP), Image Video and Multi-Dimensional Signal Processing.	
Bachelor in Engineering: Major in Mech. Engineering and minor in Computer Science	Aug 2015 – Aug 2019
KLE Technological University	<i>Hubballi, India</i>

PROJECTS

Dhan-UI: Multimodal Hands-Free Surgical Robot Interface	Siemens - FAU BEST Hackathon
- Implemented a vision pipeline using MediaPipe to track and interpret hand gestures, converting hand's spatial movements into string-based context for the Orchestrator LLM. - Integrated Google Speech API to transcribe real-time voice commands, providing a parallel speech-to-text input stream to ensure redundancy in hands-free operation. - Designed a central Orchestrator LLM using few-shot prompting to synthesize dual-channel inputs into structured, validated micro instruction commands.	
Transformers Solving the Traveling Salesman Problem	USI Deep Learning Lab
- Developed a custom Transformer model to optimize path cost for the traveling salesman problem. - Conducted statistical hypothesis testing to validate performance gains over conventional greedy algorithms. - Designed and executed hyperparameter tuning experiments to optimize model performance.	
Robotic Environment Identification and Obstacle Avoidance	USI IDISA
- Developed custom MobileNetV2 CNN model for real-time environment identification for MyT robot. - Tested various models like ResNet, YOLO for faster and leaner identification. - Implemented obstacle avoidance-based random walk algorithm for robot motion planning in ROS2. - Built multiple environments in CoppeliaSim for robot testing and simulations.	

HONOURS AND AWARDS

Bosch Automation Hackathon - First Prize	Jan 2021
Won the Bosch Automation Hackathon for the presentation of a computer vision based CAD drawing analyzer and report generator	Bosch Global Software Technologies
Bravo Award	Dec 2021
Presented for high effort, skillful contribution towards the ESP project	Bosch Global Software Technologies

LANGUAGE SKILLS

Language - English: Advanced Fluency (C1), German: Elementary Fluency (A2), Kannada, Hindi